

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 30

Annexure A

Dry Run Evaluation

Code of Conduct:

I G. Pallavi (192311208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pallavi

25 + 25 + 23 + 12 + 9
= 94
14

Student 6: GUDE PALLAVI | Register Number: 192311208

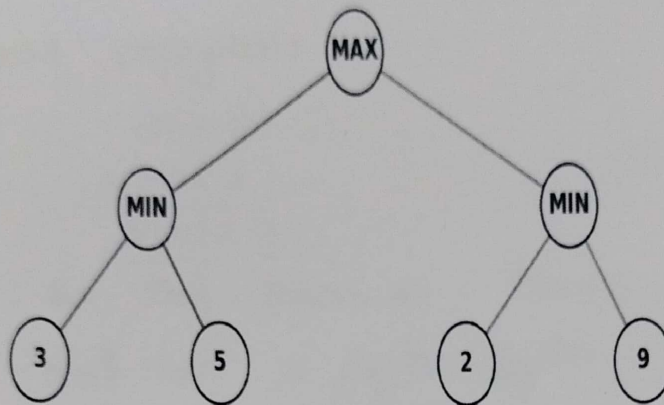
1. Question 16 - Cryptarithmic CSP

SEND + MORE = MONEY. Each letter represents a unique digit from 0-9 and M cannot be 0. Perform constraint propagation, find a valid assignment and verify the addition.

2. Question 17 - Minimax Dry Run

Perform the Minimax algorithm on the displayed game tree.

Calculate the value of each MIN node, the MAX root value and the move selected by MAX.



Node	Children Values	Value
1		
2		
3		
4		
5		

3. Question 18 - Minimax with Deeper Tree

Consider a MAX root with two MIN children. Each MIN child has two MAX children. The leaf values from left to right are: 3,5 | 2,9 | 4,6 | 1,7.

Perform a complete Minimax dry run and identify the optimal move.

Node	Children Values	Value
1		
2		
3		
4		
5		
6		
7		

Cryptarithmic CSP:-

Problem :-

$$\text{SEND} + \text{MORE} = \text{MONEY}$$

⇒ Each letter represents a unique digit from 0-9

and $M \neq 0$.

Constraint propagation:-

$$\begin{array}{r} \text{SEND} \\ + \text{MORE} \\ \hline \text{MONEY} \end{array}$$

From the ~~ten~~-thousands Column:-

• The result has 5 digits, so $M=1$

→ Using the column constraints and ensuring that all letters have different digits, we obtain:

Letter	Digit
S	9
E	5
N	6
D	7
M	1
O	0
R	8
Y	2

$$\therefore \text{SEND} = 9567$$

$$\text{MORE} = 1085$$

$$\text{MONEY} = 10652$$

Verification:-

$$\begin{array}{r} 9567 \\ + 1085 \\ \hline 10652 \end{array}$$

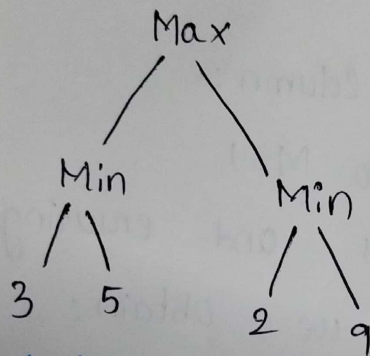
Since :- $9567 + 1085 = 10652$

the assignment is valid

$\therefore S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2$

Minimax Dry Run :-

The tree is:



Minimax Calculation:-

Node	Children values	Value
1(Max)	5, 2	5
2(Min)	3, 5	3
3(Left)	-	3
4(leaf)	-	5
5(MIN)	2, 9	2

$\therefore$ at the root :

$$\text{Max}(3, 2) = 3$$

Left Min value = $\min(3, 5) = 3$

Right Min value = $\min(2, 9) = 2$

Root Max value = $\max(3, 2) = 3$

Optimal move = left branch

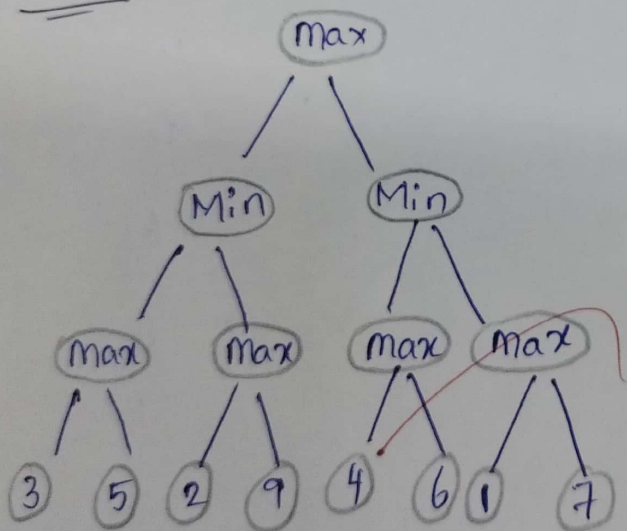
So, Max selects the left MIN node, giving a final value of 3.

3. Minimax with Deeper Tree :-

Leaf values from left to right are :-

3, 5 | 2, 9 | 4, 6 | 1, 7

Tree Structure :-



Calculation :-

Left Side :-

Node 3 = $\max(3, 5) = 5$

Node 4 = $\max(2, 9) = 9$

Node 2 = $\text{MIN}(5, 9) = 5$

Right Side :-

Node 6 = $\text{Max}(4, 6) = 6$

Node 7 = $\text{Max}(1, 7) = 7$

Node 5 = $\text{Min}(6, 7) = 6$

$\therefore$ Root = $\text{Max}(5, 6) = 6$

Complete Minimax Dry Run :-

Node	Children values	value
1. (Max)	5, 6	6
2. (Min)	5, 9	5
3. (Max)	3, 5	5
4. (max)	2, 9	9
5. (Min)	6, 7	6
6. (max)	4, 6	6
7. (max)	1, 7	7

$\Rightarrow$ Left MIN branch = 5

$\Rightarrow$ Right MIN branch = 6

$\Rightarrow$ Root max value = 6

$\Rightarrow$ Optimal move = right branch